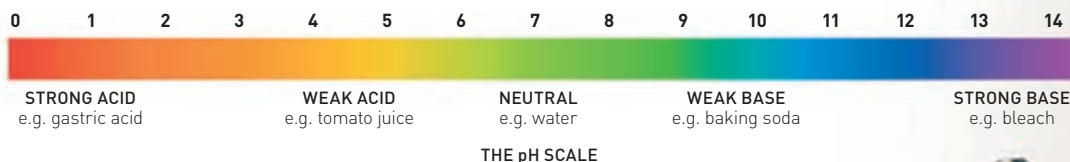


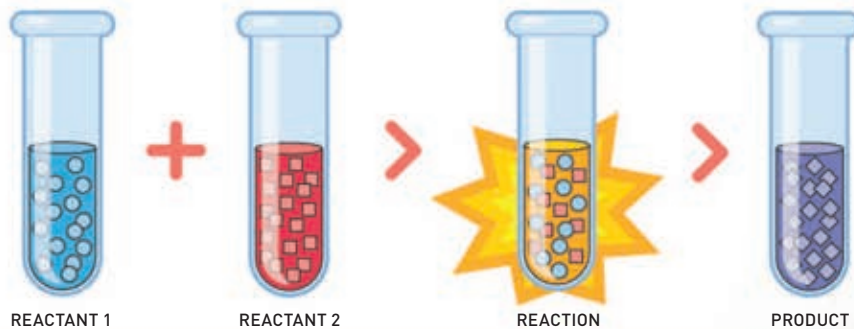
ACIDS AND BASES

All liquids and solutions fall somewhere on the acids and bases scale, which is measured as a pH level. Those at each end of the scale are very reactive and dangerous.



WHAT IS A CHEMICAL REACTION?

In the natural world, atoms and molecules are constantly joining together or breaking down to form new things. This can also be done in a laboratory. When scientists add one ingredient (called a reactant) to another, they create a chemical reaction. The molecules of the reactants split apart, rearrange themselves, and then form a new bond – the product of the reaction.



SWIRLS AND FUMES
The product of a chemical reaction can be very different from the original reactants. This mix reacts quickly, swirling and giving off fumes.

TYPES OF CHEMICAL REACTION

Although the product of a chemical reaction is very different from the reactants, none of the atoms are destroyed – there are the same number before as after the reaction. There are three types of chemical reaction.

SYNTHESIS REACTION
Two or more reactants join together to make a new compound.

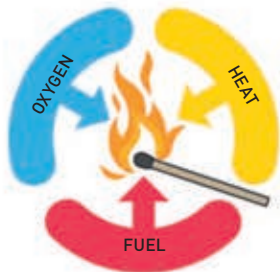
DECOMPOSITION REACTION
One reactant breaks apart into two products to make two compounds.

REPLACEMENT REACTION
Atoms of one type swap places with those of another to make a new compound.

GRAPHITE CAN BE CHANGED INTO DIAMOND THROUGH HEAT AND PRESSURE

COMBUSTION

Car engines and power stations are powered by a chemical reaction called combustion (burning). The reactants are fuel, such as petrol or coal, and oxygen from the air. Adding heat (setting fire to the fuel) starts the reaction.



MATERIALS

The materials we use for making everyday objects need to have the right properties for the object's function. For example, wood is robust and good for building a chair but would be a poor choice for a frying pan, because it would catch fire.



WOOD
Hard, strong, and rigid, burns readily, and is a good insulator.



METAL
Good conductor of heat and electricity. It is strong and inflexible.



PLASTIC
Strong, waterproof, and can be made into any shape. Good insulator.



CERAMIC
Fragile if knocked but can withstand high temperatures.

GREAT CHEMISTS

The discoveries of great chemists have contributed to human progress in everything from medicine to space travel.

- **ROBERT BOYLE (1627–91)**
The author of *The Sceptical Chymist* was the first to develop rigorous scientific techniques for his experiments in the field of chemistry.
- **ANTOINE LAVOISIER (1743–94)**
The first chemist to demonstrate that water is made of oxygen and hydrogen, and to show that oxygen is needed for combustion.
- **MARIE CURIE (1867–1934)**
Twice winner of the Nobel prize, Curie discovered radium.
- **LINUS PAULING (1901–94)**
American scientist who worked out how molecules bond together.
- **DOROTHY HODGKIN (1910–94)**
A pioneer in X-ray techniques who discovered the atomic structure of penicillin.

LINUS PAULING



GLASS
Transparent and can be made into any shape. Breaks easily if thin.



SYNTHETIC FIBRE
Plastic-based fibres are strong and waterproof.



KEVLAR®
High-strength material that withstands high impact and extremes of temperatures.